Yilin “Harry” Lu

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Research Interests

My research interests centered on human-AI collaboration, explainable AI, and interactive visualization systems. Specifically, I design and develop better visualization techniques and interactive systems to help people learn, understand, and steer AI models and mechanisms.

Education

University of Minnesota, Twin Cities, B.A.(with Honors) in Computer Science, Mathematics, Economics(Quantitative-emphasis), and Minor in Statistics — Sept 2023 – May 2027

- **CGPA**: 3.8/4.0
- **Awards**: UROP Scholarship - Fall 2024($1800), Dean’s Lists(2023, 2024, 2025), University Honors Program
- **Graduate Coursework(5000 and above):** Visualization for Interpretable AI, Visualization, Machine Learning for Health, Bayesian Statistics for Astronomy, Cryptography and Number Theory(Fall 2025)

Cornell University, Winter Semester in Network Economics — Jan 2025

- GPA: 4.3/4.3(A+)

San Diego State University, Dual Enrollment in Mathematics — Sept 2022 - May 2023

Torrey Pines High School, High School Diploma — Spet 2021 - May 2023

Featured Research Works

[P3] The Agentopia Framework: Understanding LLMAS Mechanisms through a Game Design Framework — Oct 2025

Yilin Lu, Shurui Du, Qianwen Wang

ACM IUI 2026 - Full Paper Submitted

[C1] The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay — July 2025

Yilin Lu, Shurui Du, Qianwen Wang

IEEE Visualization Conference (VIS) 2025

[P2] GNN 101:Visual Learning of Graph Neural Networks in Your Web Browser — Sept 2025

Yilin Lu, Chongwei Chen, Yuxin Chen, Kexin Huang, Marinka Zitnik, Qianwen Wang

IEEE TVCG Journal (Fast-tracked) - Submitted

[W1] What Can a Node Learn from Its Neighbors in Graph Neural Networks — Oct 2024

Yilin Lu, Chongwei Chen, Matthew Xu, Qianwen Wang

IEEE Visualization Conference (VIS) 2024 - the 7th VISxAI Workshop

Other Research Works

[P1] Optimizing the Tool Calling Capabilities of Large Language Models Using In-Context Learning — Aug 2025

Jiaqi You, Yusheng Ma, **Yilin Lu**, Wei Zhu, Maria Gini

Neurocomputing Journal - Full Paper Submitted

[R1] GRB231115A: TURBO Pre-Burst Optical Upper Limits — Dec 2023

R. Strausbaugh, Daniel Warshofsky, Pat Kelly, Mandeep S. S. Gill, Alexandre Toscano, **Yilin Lu**, Sydney Leggio, Aksinya Kamenshikova

NASA General Coordinate Networks

Research Talks

[T4] The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay Nov 2025

Yilin Lu *IEEE VIS 2025 (Upcoming!)*

[T3] The Agentopia Times: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay Aug 2025

Yilin Lu, Shurui Du *IEEE VIS 2025, UMN CS&E Graduate Student Orientation Poster Session (Invited)*

[T2] GNN 101:Visual Learning of Graph Neural Networks in Your Web Browser Oct 2024

Yilin Lu *UMN Visual Computing Seminar (Invited)*

[T1] What Can a Node Learn from Its Neighbors in Graph Neural Networks Oct 2024

Yilin Lu *the 7th VISxAI Workshop at VIS 2024*

Experience

Research Team Lead, the Department of CS&E, Univ. of Minnesota – MN, USA April 2024 – Present

- **Topic**: Designing novel systems for human-centered AI and explainable AI, supervised by Prof. Qianwen Wang.
- Lead a team of 6 to develop and maintain 3 R&D projects responsible for project and team management.
- Wrote 3 scientific papers to communicate the scientific ideas, methodologies, evaluation, and results.
- Led 2 revision report writings for conference/journal revision.

Science Workflow Intern, Minnesota Institute for Astrophysics – MN, USA Sept 2023 – May 2024

- **Topic**: Designing crowdsourcing systems for astronomy, supervised by Prof. Pat Kelley and Dr. Mandeep.
- Developed a crowd-sourcing website for the building of astronomical datasets in Zooniverse Project Builder.
- Built a data streamline software to enables the real-time I/O between telescopes and remote servers.
- Engineered a data processing pipeline for asteriod images in OpenCV and SciPy.

VR Engineering Intern, Beijing Dream Blossom Technologies – Remote July 2022 – Oct 2022

- Implemented a Virtual Reality UI System on the Oculus platform using Unity 3D.
- Built 3D character models and used skeleton components and inverse kinematics to create the animations and controls of the character in Unity 3D and Blender.

Research Prototypes

Aligning Human-AI Collaborative Writing for Scientific Literature Link

- Developed a simulation pipeline for LLM scientific iterative writing with various configurations.
- Engineered a computational data pipeline to analyze the alignment of human-AI cognitive processes.
- Deployed visual analytical tool to explore the iterative scientific writing process.
- Tools Used: Python, pymc, Transformers, HuggingFace, TypeScript, D3.js, Svelte.

Undergraduate Research Projects in Economics and Mathematics

[UG1] Partial Orderings and Linear Orderings: An Investigation of Hausdorff Maximal Theorem and Linear Extension Theorem Fall 2025

Yilin Lu *Mathematics Capstone Research Paper*

[UG2] Digital Development Convergence in the United States: A Spatial Temporal Analysis from 2013-2023 Aug 2025

Yilin Lu *UG Economic Research Paper - Highest Scored Paper in ECON4331W*

Research Mentorship

Shurui Du, B.Sc., Computer Science - Univ. of Minnesota, Twin Cities Feb 2025 – Present
Chongwei Chen, B.A., Mathematics - Univ. of Minnesota, Twin Cities May 2024 - Sept 2024

Press

Visual Computing Center Film Summer 2025

UMN CSE Press, Visual Computing Center at University of Minnesota

GNN 101 featured in CS224W: Learning on Graphs Fall 2024

Stanford University, CS224W: Learning on Graphs

Personal Projects

Game Portfolio(Game Design and Development) Link

- Worked as a full-stacked game developer participated various hackathons(e.g., Ludum Dare, Global Game Jam), highest ranking reached top 25%.
- Responsible for art design, gameplay design, game programming, and post-development maintenance.
- Tools Used: Unity 3D, Gamemaker Studio 2, Godot Engine, Aseprite, Photoshop.

Zhihu Blog Link

- Write posts about research in HCI, game technology, and computer programming, reached 1900+ followers.

Online Community Moderation (2018-2022) Link

- Moderated an online community with 7.5k+ members, under my tenure, membership has grown by 3k users.
- Organized two print-and-play (PNP) game design competitions through forum (2018 and 2020).

Volunteering Experience

Community Engagement Volunteer, UMNTC& Upstream Arts – Minneapolis, MN Sep 2023 – Dec 2023

Technologies

Languages: Python, TypeScript, R, C/C++

Technologies: Numpy, Pandas, Matplotlib, statsmodels, PyTorch, Scikit-learn, Transformers, Huggingface, OpenCV, LangChain, LangGraph, D3.js, React, Three.js, Flask, Unity, Godot, Phaser, Tidyverse, ggplot2